Synthetic Is All You Need?

Lior Vinman *‡
vinmanl@post.bgu.ac.il

Jonathan Zmora *‡
zmoraj@post.bgu.ac.il

Abstract

Transferring computer vision algorithms from simulated environments to real-world applications remains a significant challenge due to large domain gaps caused by lighting, textures, sensor noise, and other physical inconsistencies. In this work, we tackle the synthetic-to-real generalization problem of chessboard state classification. First, we address the visual domain gap between synthetic and real chessboard images by engineering a highly realistic synthetic dataset via a 3D rendering pipeline using Blender. Training on this high-fidelity dataset yields significant zero-shot performance gains over a naively rendered synthetic dataset when evaluated with a ResNet18 baseline model. For our primary zero-shot evaluation, we propose an architecture utilizing a local per-square ConvNeXt feature extractor coupled with a Transformer Encoder to capture global board context. To adapt this knowledge to real-world images, we transition to a transfer-learning approach, fine-tuning the ConvNeXt backbone's final stage with a classification head on a limited set of real-world images. Across both phases, we integrate an Integer Linear Programming (ILP) solver as a fundamental architectural component to enforce strict chessboard state constraints. This solver successfully rectifies illegal predictions, driving our final model to a 97.3% accuracy on the real-world test dataset.

*Equal contribution. Listing order is random.

‡The Stein Faculty of Computer and Information Science, Ben-Gurion University of the Negev, Israel; 1st-year M.Sc. Computer Science students.

1 Introduction

Given a single RGB image of a physical chessboard, the primary objective of this task is to classify each of the 64 individual board squares into one of the possible piece classes or identify it as an empty square. The system must then utilize these per-square predictions to output the corresponding board state, commonly represented in Forsyth-Edwards Notation (FEN), and construct the complete chessboard diagram.

Digitizing over-the-board chess games is crucial for game analysis, broadcasting, and archival purposes. However, training robust deep neural networks to perform this task requires massive amounts of labeled data. Manually annotating the exact class and location of up to 32 pieces across thousands of chessboard images is prohibitively expensive and prone to human error. Therefore, utilizing procedurally generated synthetic data is a highly attractive alternative.

However, models trained exclusively on naively generated synthetic data typically suffer from a steep performance drop when applied to real-world images, due to the *sim-to-real domain gap*. This domain gap is based on physical phenomena absent in basic 3D synthetic renderings: variable room lighting, specular highlights, realistic textures, sensor noise, poor image quality, and other physical characteristics.

Another fundamental challenge lies in the formulation of the problem. Treating the problem as a single whole-board classification task involves a very large output space, rendering direct network optimization impractical. In contrast, decomposing the problem into 64 isolated per-square classification tasks drastically simplifies the optimization process to a standard 13-class problem. However, this isolated approach discards critical global board context, ignoring the inherent structural rules of chess and the spatial relationships between pieces that could otherwise aid the network in resolving ambiguous visual features.

Our goal is to successfully bridge the sim-to-real domain gap by curating a highly realistic synthetic training dataset, while developing a neural network architecture capable of leveraging both localized per-square features and the global spatial context of the board.

The main contributions of this work are as follows:

- **Synthetic Data Rendering Pipeline:** We engineered a highly realistic synthetic chessboard dataset using an extensible and adaptable 3D Blender rendering pipeline that accurately models physical phenomena inherent to real-world environments. We make this generation pipeline publicly available to facilitate future research, and demonstrate that this data-centric approach significantly improves zero-shot transfer performance compared to low-fidelity synthetic datasets.

- **Hybrid Architecture:** We propose a hybrid neural network architecture that uses a ConvNeXt local feature extractor coupled with a Transformer Encoder, empirically demonstrating that the integration of global spatial board context improves zero-shot chessboard state classification. Furthermore, we show that this robust synthetic foundation enables highly efficient domain adaptation; fine-tuning only the final stage of the ConvNeXt backbone and a linear classification head on a limited set of real-world images is sufficient to achieve exceptional real-world performance.

- **Logically Constrained Inference:** We formulate and integrate an Integer Linear Programming (ILP) solver as a final logical constraint layer. By mathematically enforcing strict chessboard state constraints, the solver successfully rectifies illegal board state predictions, driving our final model to a test accuracy of 97.3%.

2 Related Work

2.1 Chessboard State Classification

The task of digitizing chessboard states from images has been explored through various traditional and deep learning paradigms. Wölflein and Arandjelović (2021) achieved high classification accuracy on purely synthetic datasets. Although demonstrating great results, synthetic chessboard state classification is a fundamentally easier task than real chessboard state classification. Masouris et al. (2024) attempted to train end-to-end models for real chessboard state classification directly on real-world chessboard images. However, this approach is limited by the immense volume of labeled data needed for a robust solution. Our work directly addresses this annotation bottleneck. We approach the problem through a sim-to-real lens, focusing on generating highly robust synthetic data that transfers to the real world without requiring massive real chessboard image datasets.

2.2 The Sim-To-Real Gap

Generating synthetic data to train models for physical-world deployment is a cornerstone of modern computer vision. The foundational work on Domain Randomization by Tobin et al. (2017) demonstrated that neural networks can generalize to real-world environments by training on simulated images with heavily randomized textures and lighting. While their approach proves that the "reality gap" can be bridged using simulated data, it relies on massive visual diversity to force the network to ignore the synthetic nature of the training set. Our work differentiates itself by focusing on targeted physical traits rather than pure randomization. By utilizing an extensible 3D Blender pipeline to render high-fidelity physical phenomena specific to chessboards such as specular highlights on wooden pieces and grain in a wooden chessboard, we demonstrate that realistic, physically grounded rendering is crucial in zero-shot transfer tasks.

2.3 Logically Constrained Neural Networks

Many contemporary approaches attempt to teach neural networks the rules of their environment implicitly through customized loss functions or by embedding differentiable optimization layers directly into the training loop (Amos and Kolter, 2017). However, these implicit, gradient-based methods rarely guarantee absolute compliance with rigid structural rules. Our approach differs by strictly decoupling the visual feature extraction from the logical constraints. By formulating the structural rules of a chessboard as an Integer Linear Programming (ILP) problem, we apply logic as mathematical constraints on top of the network's output logits. This guarantees legal board configurations and resolves visual ambiguities without requiring the network to implicitly learn the rules of the game.

3 Methodology

3.1 Dataset Generation and Composition

The foundation of our approach is a robust, large-scale synthetic dataset designed to capture a wide variance of chess board states. To ensure comprehensive coverage of possible board configurations, we aggregated roughly 3,700 FEN strings from three distinct sources. First, we utilized a baseline set (which, was provided by the TA) of approximately 500 FENs extracted from five standard games. To introduce high-level positional complexity and master-level structures, we acquired approximately 1,200 FENs parsed from the Portable Game Notation (PGN) files of Grandmaster Nodirbek Abdusattorov's historical matches. Finally, to prevent the model from overfitting to standard opening theory and human-logical board states, we developed a custom Python script utilizing the `python-chess` library to simulate random legal games, generating an additional 2,000 highly randomized FEN configurations.

These FEN strings were fed into a customized Blender rendering pipeline to produce 3D synthetic images of the corresponding boards. To capture distinct geometric and lighting features, each FEN was rendered from three specific camera angles: "Overhead" (a pure top-down perspective), "West" (angled from the right), and "East" (angled from the left). This yielded three distinct full-board images per FEN configuration.

3.2 Preprocessing and Class Definition

The full-board synthetic images, initially rendered at a resolution of $P \times P$ pixels, were systematically cropped into 64 individual squares. Each extracted cell was subsequently resized to a standardized input resolution of 64×64 pixels and normalized using standard ImageNet channel means and standard deviations.

Our classification task defines 13 mutually exclusive target classes: one representing an `empty` square, and 12 representing the specific piece types differentiated by color (`white_pawn`, `black_pawn`, `white_knight`, ..., `black_king`). Processing the full dataset across all camera angles yielded a massive corpus of approximately 710,400 individual cell images. This dataset was subsequently partitioned into training ($Y\%$), validation ($Z\%$), and testing ($X\%$) splits to strictly isolate evaluation data.

3.3 Model Architecture and Training Protocol

We formulated the cell-by-cell recognition problem as a standard image classification task, employing a ResNet18 convolutional neural network backbone. The standard fully connected layer of the ResNet18 was modified to output logits for our 13 target classes.

The network was trained using a supervised learning paradigm. During training, we monitored both loss and accuracy across epochs, saving the model weights that achieved the highest validation accuracy to prevent overfitting. As detailed in our experimental logs, the training loop dynamically adjusted the learning rate upon reaching plateaus in validation loss, ensuring smooth convergence as the model learned to extract abstract piece features rather than memorizing synthetic textures. We used the Adam optimizer with a learning rate of 0.001 and Cross-Entropy Loss.

4 Experiments and Results

To systematically address the synthetic-to-real domain gap, we conducted a series of ablation studies, iterating upon our baseline model. For all experiments, performance was evaluated on a strictly isolated test set of real-world chessboard images. The primary metric is square-level classification accuracy (out of 13 classes).

4.1 Baseline Synthetic-to-Real Evaluation

The initial baseline model was trained purely on unaugmented synthetic data. As expected, the ResNet18 quickly memorized the pristine synthetic textures, achieving an over-confident 99.99% validation accuracy on the synthetic validation split. However, when deployed on the real-world test set, the square-level classification accuracy dropped significantly to 64.33%. This steep degradation empirically demonstrates the severity of the domain gap. Furthermore, the strict full-board accuracy (where all 64 squares must be correctly classified simultaneously) remained at 0.00%, highlighting how highly sensitive full-state reconstruction is to isolated single-square misclassifications.

4.2 Experiment 1: Mitigating Class Imbalance

A fundamental characteristic of standard chess games is the heavy prevalence of empty squares relative to occupied squares, particularly in the endgame. To prevent the model from developing a statistical bias toward predicting the `empty` class, we integrated Weighted Random Sampling into the training DataLoader. By artificially re-balancing the sampling probability of rare pieces (e.g., queens and kings) versus empty squares, the model learned more robust features for minority classes. This modification improved our real-world square accuracy to 65.26%.

4.3 Experiment 2: Spatial Context and Cropping Padding

We originally hypothesized that incorporating neighboring spatial context around each extracted cell might aid the model in classification—for example, by allowing the network to see the bases of tall pieces that might bleed into adjacent squares. To test this, we evaluated varying cell-crop padding factors ranging from 0.0 (strict grid crop) to 1.5 (heavy overlap).

Surprisingly, the empirical results directly contradicted this hypothesis. The strictly constrained crop (padding of 0.0) yielded the highest accuracy in this ablation (64.88%). Introducing context generally degraded performance, dropping to 61.65% at a 0.2 padding factor and settling at 62.84% for a 1.5 padding factor. We conclude that capturing overlapping pieces from adjacent squares acts as confounding noise rather than helpful context for the ResNet18.

4.4 Experiment 3: Data Augmentation for Domain Adaptation

Because the Blender-generated FEN renders lack the physical imperfections of real-world photography, we introduced a pipeline of stochastic data augmentations to simulate real-world camera artifacts. By applying varying degrees of blur, random noise, and color jitter during the training phase, we forced the network to disregard clean synthetic textures and rely entirely on robust structural and geometric features. This domain adaptation strategy proved highly effective, culminating in our best-performing model, which achieved a peak square-level accuracy of 66.89% on the unseen real-world dataset.

5 Conclusion and Future Work

In this paper, we addressed the complex challenge of chessboard state recognition by focusing on the synthetic-to-real domain gap. To overcome the scarcity of annotated real-world datasets, we developed an automated pipeline to render 3D chessboard images from FEN strings using Blender, generating a training corpus of over 700,000 isolated square images. We demonstrated that while a baseline ResNet18 classifier easily memorizes clean synthetic textures, it fundamentally struggles to generalize to the messy reality of physical photography.

Through systematic ablation studies, we proved that naive spatial padding degrades performance by introducing confounding adjacent-piece noise. However, by leveraging weighted random sampling to combat class imbalance (specifically the over-representation of empty squares) and applying aggressive stochastic data augmentations to simulate real-world camera artifacts, we successfully improved the model's robustness. Our final optimized model achieved a peak square-level classification accuracy of 66.89% on an unseen real-world dataset.

Despite these improvements, achieving perfect full-board reconstruction remains a highly sensitive challenge, as a single misclassified square out of 64 completely invalidates the overall board state (resulting in a 0.00% full-board accuracy). To bridge this final gap, we propose three distinct avenues for future work:

- **Algorithmic Post-Processing:** Integrating a chess-logic rules engine to validate and correct predictions. By enforcing standard game rules (e.g., prohibiting multiple kings of the same color, validating legal pawn structures), the system could automatically fall back to the model's second-most-likely class prediction when an illegal board state is detected.
- **Hybrid Fine-Tuning:** Utilizing our synthetically trained model as a strong baseline prior and performing few-shot fine-tuning on a small, manually annotated real-world dataset. This hybrid approach is a standard and highly effective method for overcoming the final threshold of a domain gap.
- **Contextual Architectures (Vision Transformers):** Since our experiments indicated that expanding physical crop padding introduces detrimental noise, future iterations could replace the ResNet18 backbone with a Vision Transformer (ViT). Attention mechanisms would allow the network to evaluate the global 8×8 grid context simultaneously while classifying individual squares, bypassing the limitations of artificial cell cropping.